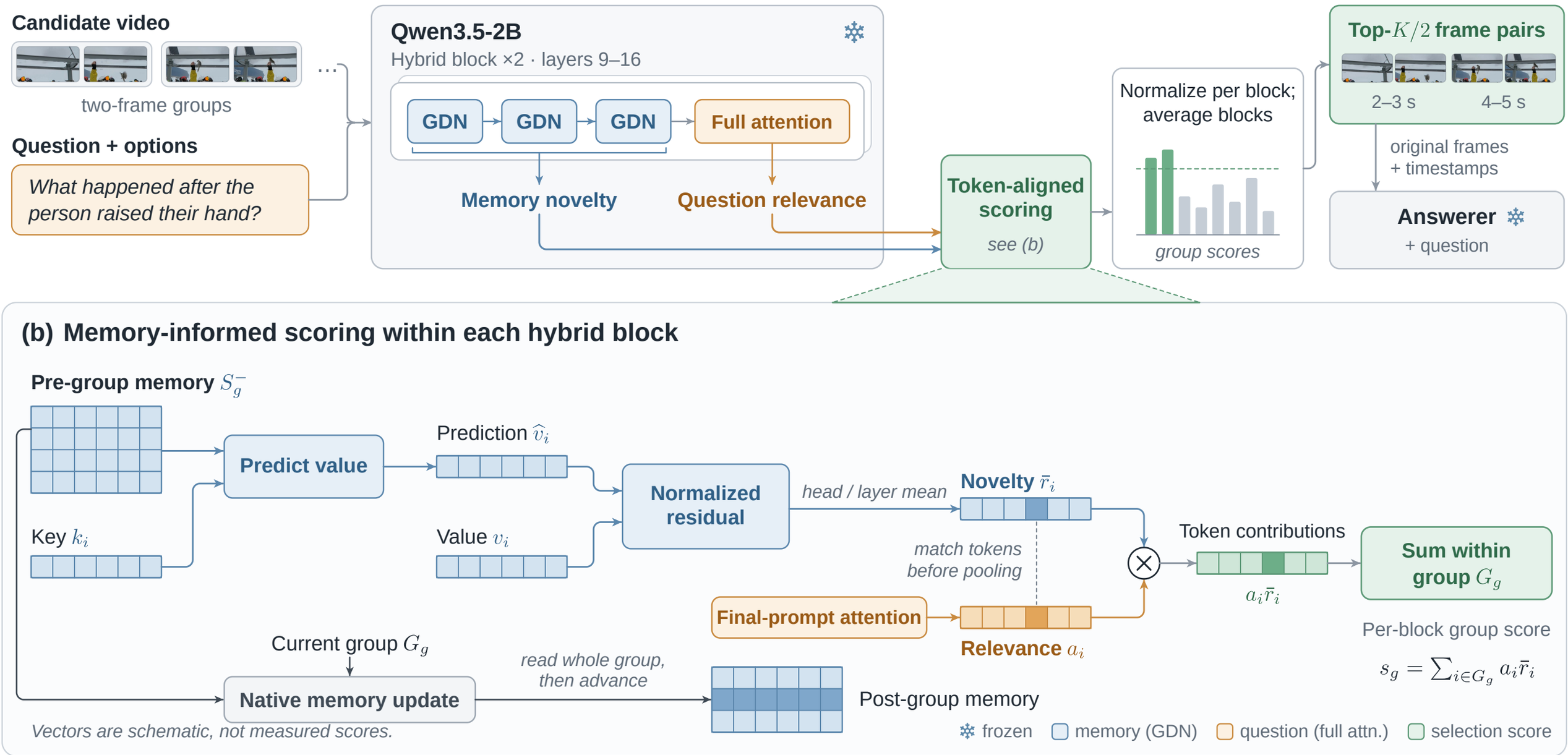

(a) Selection with a pretrained hybrid backbone
Candidate video
two-frame groups
Question + options
What happened after the person raised their hand?
Qwen3.5-2B
Hybrid block ×2 · layers 9–16
GDN
GDN
GDN
Full attention
Memory novelty
Question relevance
Token-aligned scoring
see (b)
Normalize per block; average blocks
group scores
Top-$K/2$ frame pairs
2–3 s
4–5 s
original frames + timestamps
Answerer
+ question
(b) Memory-informed scoring within each hybrid block
Pre-group memory S_g^-
Predict value
Prediction $\hat{v}_i$
Key k_i
Value v_i
Normalized residual
head / layer mean
Novelty $\bar{r}_i$
match tokens before pooling
Final-prompt attention
Relevance a_i
Token contributions
$a_i \bar{r}_i$
Sum within group G_g
Per-block group score
$s_g = \sum_{i \in G_g} a_i \bar{r}_i$
Current group G_g
Native memory update
read whole group, then advance
Post-group memory
Vectors are schematic, not measured scores.
frozen
memory (GDN)
question (full attn.)
selection score